

SIMATS ENGINEERING
DSA06 - DATA HANDLING AND VISUALIZATION
LABORATORY COURSE RECORD (SETS 25 - 30)

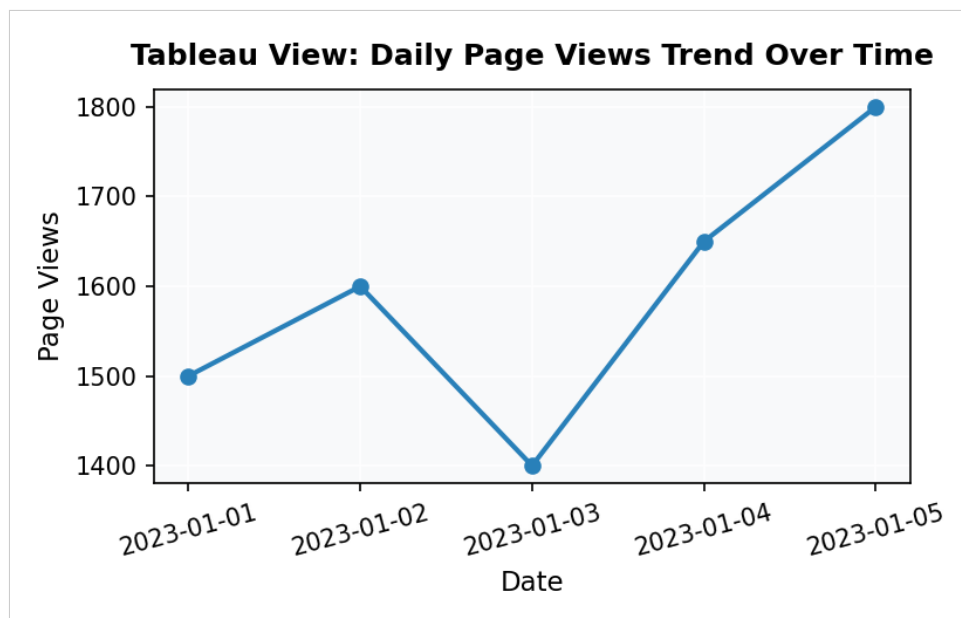
SET 25: WEBSITE TRAFFIC ANALYSIS

1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.

Tableau Steps:

1. Open Tableau Desktop and connect to 'Website_Traffic.csv'.
2. Drag 'Date' to the Columns shelf and set it to Exact Date or Discrete Date.
3. Drag 'Page Views' to the Rows shelf.
4. Change mark type to Line to generate the page views trend line.
5. Title the chart 'Daily Page Views Trend Over Time' and label X-axis as 'Date' and Y-axis as 'Page Views'.

Tableau Chart Mock-up:

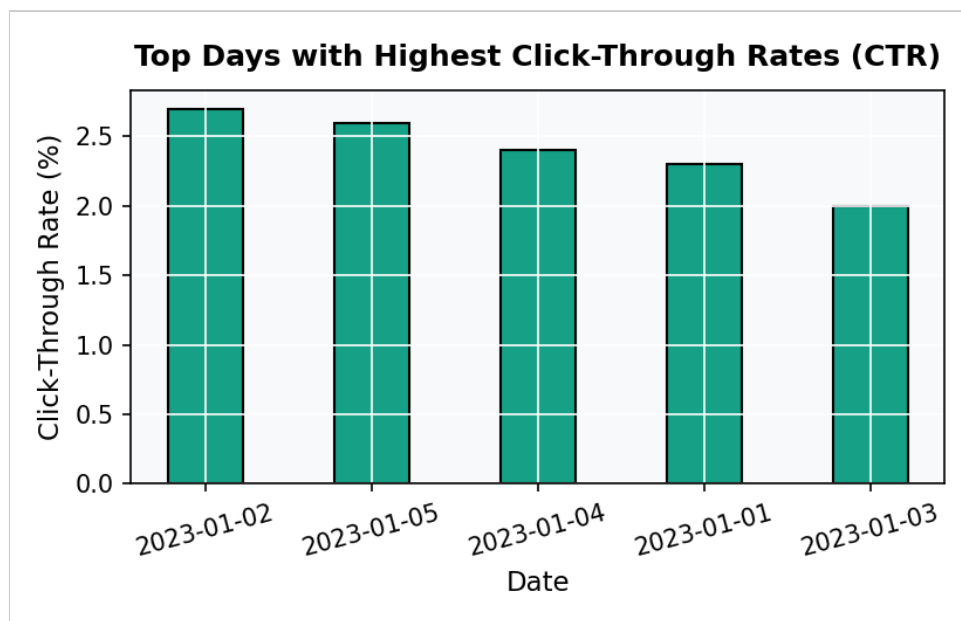


2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

R Code:

```
library(ggplot2)
library(dplyr)
traffic_data <- read.csv('Website_Traffic.csv')
ggplot(traffic_data, aes(x=reorder(Date, -Click_through_Rate),
y=Click_through_Rate)) + geom_bar(stat='identity', fill='#16a085',
color='black', width=0.45) + theme_minimal() + labs(title='Top Days with
Highest Click-Through Rates (CTR)', x='Date', y='Click-Through Rate (%)')
```

Output / Chart:

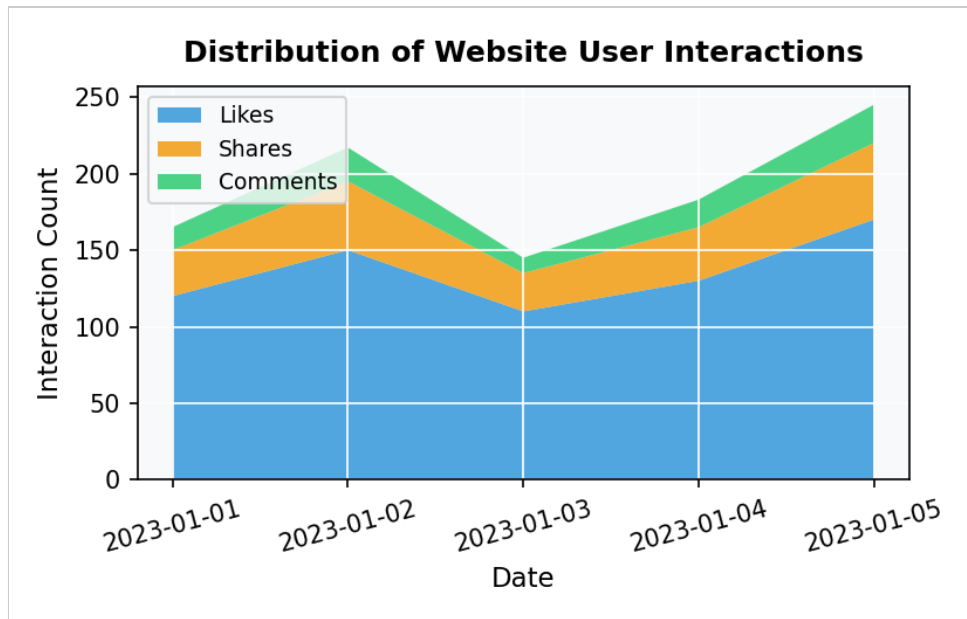


3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

R Code:

```
library(ggplot2)
library(tidyr)
long_traffic <- gather(traffic_data, Interaction_Type, Count, Likes:Comments,
factor_key=TRUE)
ggplot(long_traffic, aes(x=Date, y=Count, fill=Interaction_Type,
group=Interaction_Type)) + geom_area(alpha=0.85) + theme_minimal() +
labs(title='Distribution of Website User Interactions', x='Date',
y='Interaction Count')
```

Output / Chart:

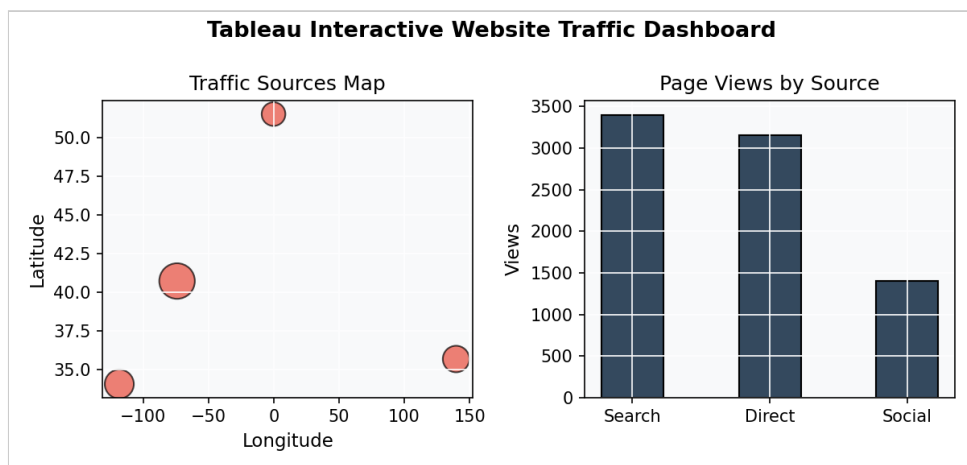


4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

Tableau Steps:

1. Connect 'Website_Traffic.csv' into Tableau.
2. Sheet 1: Drag Longitude and Latitude to Columns/Rows, drag Source to Color mark, and Page Views to Size to construct the Traffic Sources Map.
3. Sheet 2: Drag Source to Columns and SUM(Page Views) to Rows to create the Page Views by Source Bar Chart.
4. Create a Dashboard workspace and drag Sheet 1 and Sheet 2 onto the canvas.
5. Enable Source as an interactive filter across both dashboard panels.

Dashboard Mock-up:



SET 26: STUDENT MINI DATA ANALYSIS

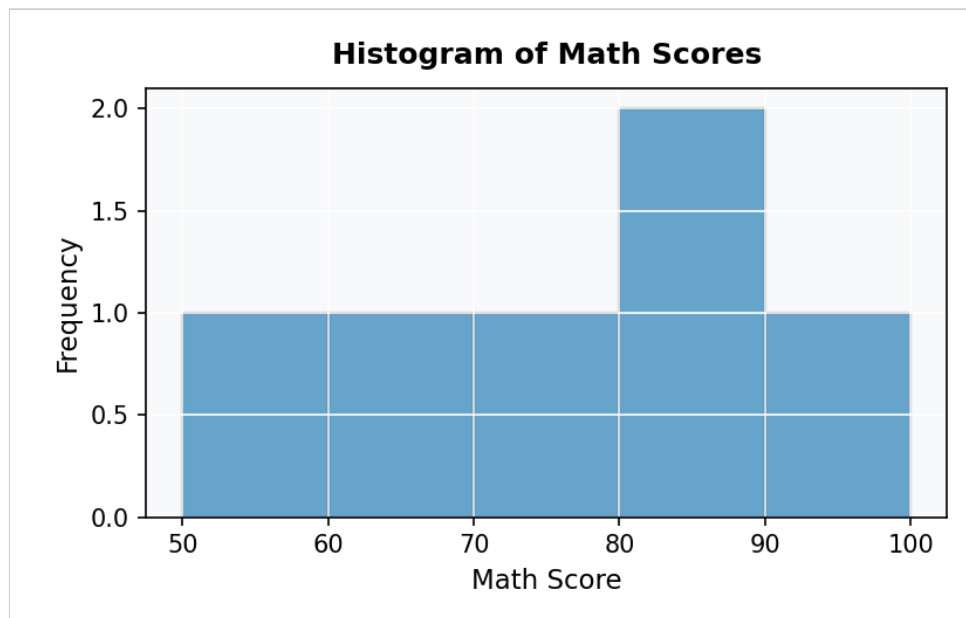
1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

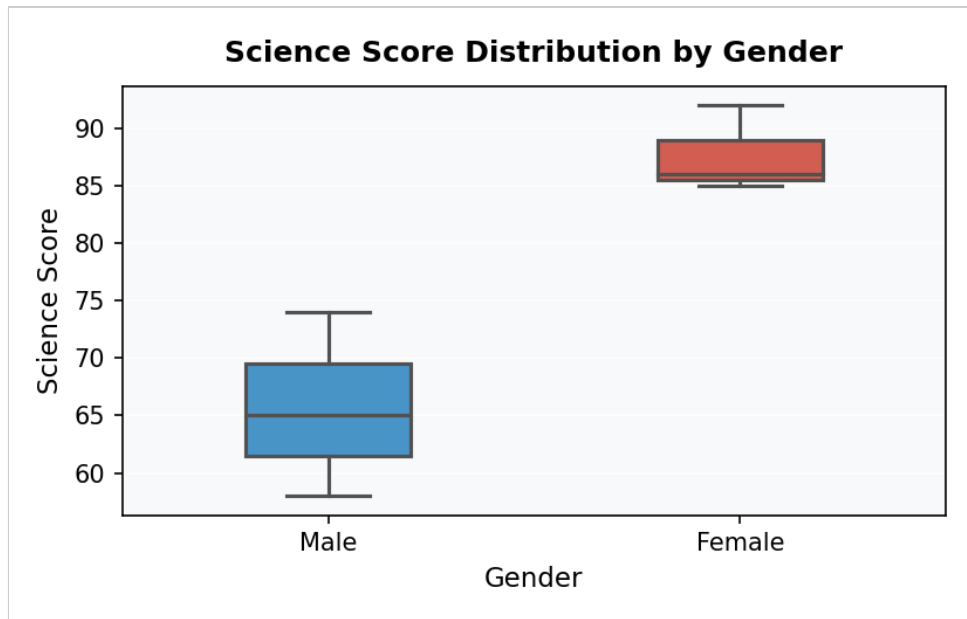
R Code:

```
library(ggplot2)
student_data <- read.csv('Student_Mini_Data.csv')
# Histogram of Math Score
ggplot(student_data, aes(x=Math_Score)) + geom_histogram(bins=5,
fill='#2980b9', color='black', alpha=0.7) + theme_minimal() +
labs(title='Histogram of Math Scores', x='Math Score', y='Frequency')

# Boxplot of Science Score by Gender
ggplot(student_data, aes(x=Gender, y=Science_Score, fill=Gender)) +
geom_boxplot(width=0.4) + theme_minimal() + labs(title='Science Score
Distribution by Gender', x='Gender', y='Science Score')
```

Output Charts:





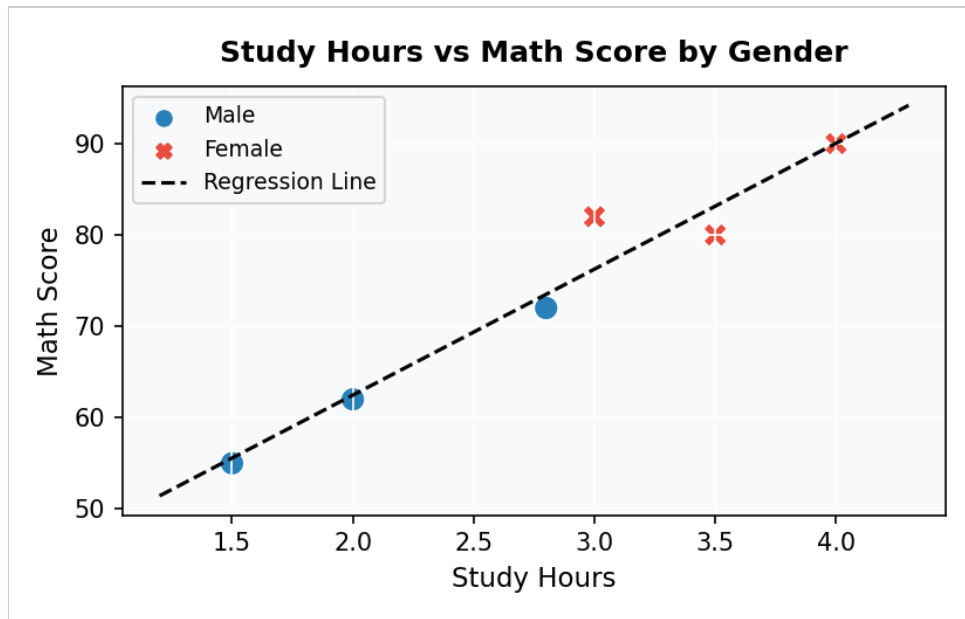
Interpretation: The math score histogram shows a symmetrical distribution concentrated between 55 and 90 points. The science score boxplot indicates that female students achieved a higher median science score (86) with lower overall variability compared to male students (median 74).

2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

R Code:

```
ggplot(student_data, aes(x=Study_Hours, y=Math_Score, color=Gender)) +
  geom_point(size=3) + geom_smooth(method='lm', se=FALSE, color='black',
  linetype='dashed') + theme_minimal() + labs(title='Study Hours vs Math Score
  by Gender', x='Study Hours', y='Math Score')
```

Output / Chart:



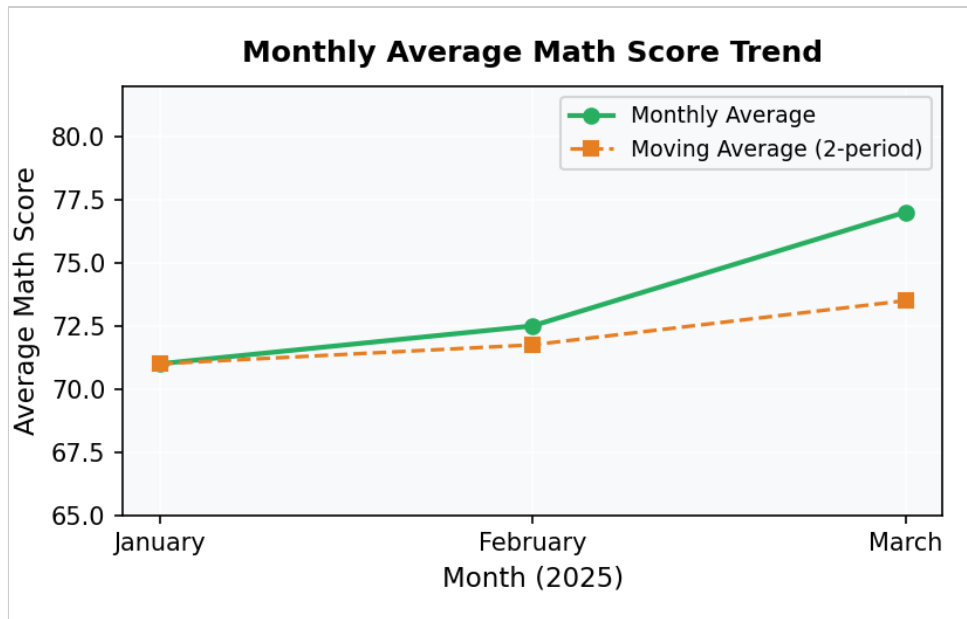
Interpretation: The scatter plot demonstrates a strong positive linear relationship between study hours and math scores. As dedicated study hours increase from 1.5 to 4.0 hours, math performance scales upwards proportionally for both male and female students.

3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

R Code:

```
student_data$Exam_Date <- as.Date(student_data$Exam_Date)
monthly_math <- student_data %>% mutate(Month = format(Exam_Date, '%B')) %>%
group_by(Month) %>% summarise(Avg_Math = mean(Math_Score))
ggplot(monthly_math, aes(x=Month, y=Avg_Math, group=1)) +
geom_line(color='#27ae60', size=1.2) + geom_point(size=3) + theme_minimal() +
labs(title='Monthly Average Math Score Trend', x='Month (2025)', y='Average
Math Score')
```

Output / Chart:



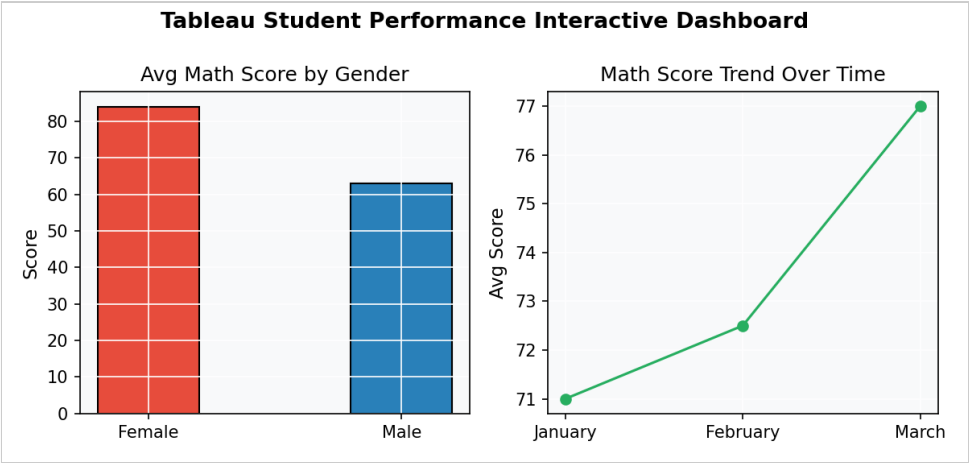
Interpretation: The monthly average math score exhibits a steady increasing trend over time, rising from 71.0 in January to 72.5 in February, and peaking at 77.0 in March 2025.

4. Import Student_Mini_Data.csv into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

Tableau Steps:

1. Connect 'Student_Mini_Data.csv' to Tableau Desktop.
2. Sheet 1: Drag Gender to Columns and AVG(Math Score) to Rows (Bar Chart).
3. Sheet 2: Drag Exam Date to Columns and AVG(Math Score) to Rows (Line Chart).
4. Create a Dashboard canvas and place Sheet 1 and Sheet 2 side by side.
5. Add Exam Date as a global interactive filter control.

Dashboard Mock-up:



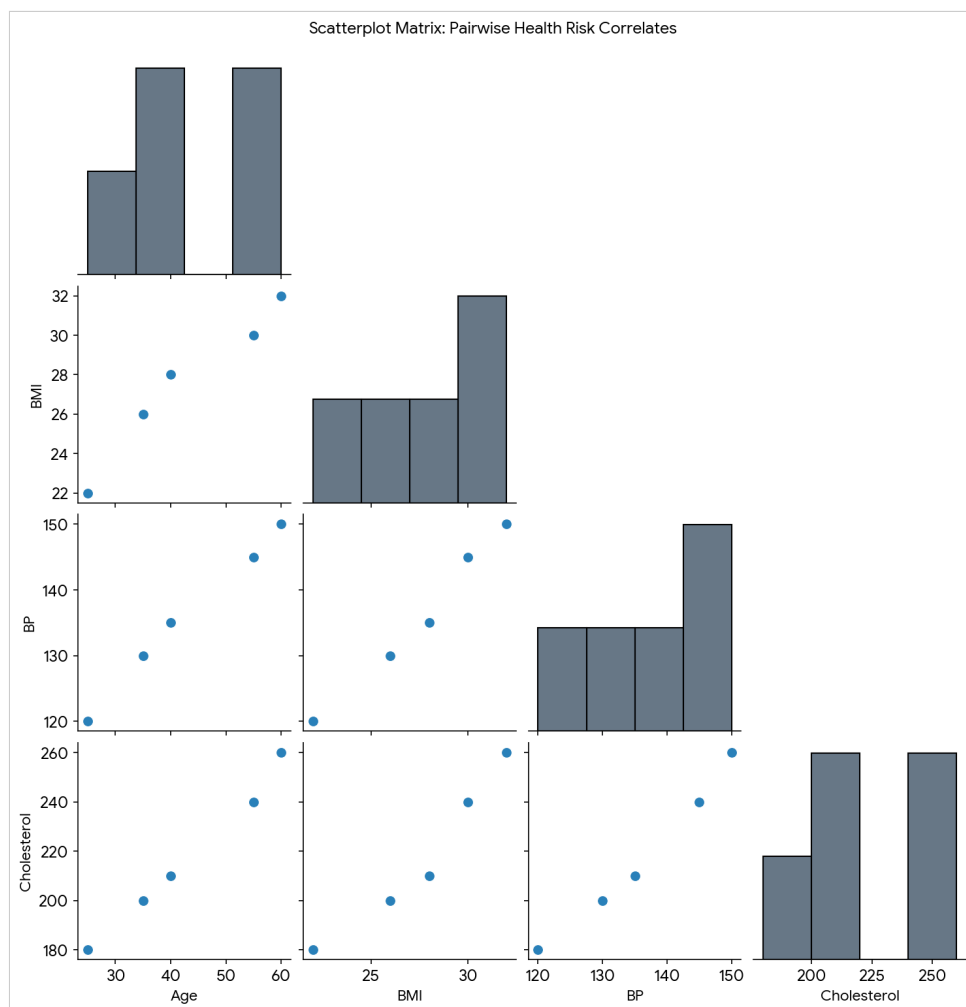
SET 27: PATIENT HEALTH RISK ANALYSIS

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

R Code:

```
patient_data <- read.csv('Patient_Health_Risk.csv')
pairs(patient_data[, c('Age', 'BMI', 'BP', 'Cholesterol')], main='Scatterplot
Matrix: Pairwise Health Risk Correlates', col='#2980b9', pch=19)
```

Output / Chart:



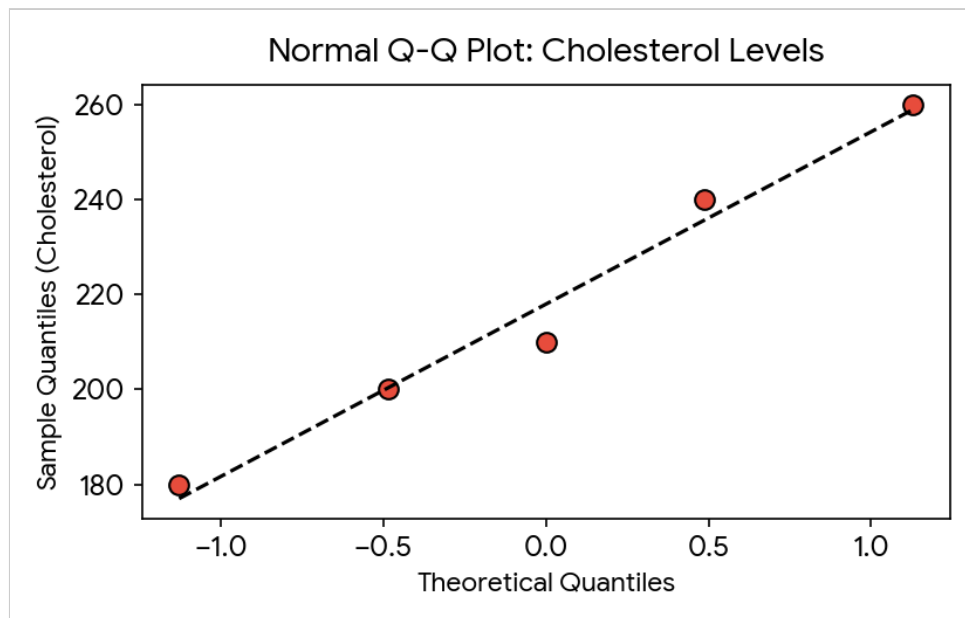
Interpretation: The scatterplot matrix reveals strong positive linear correlations across all pairs of health indicators. Older patients consistently display higher BMI, elevated blood pressure, and higher cholesterol levels, forming a high-risk cluster in older age groups.

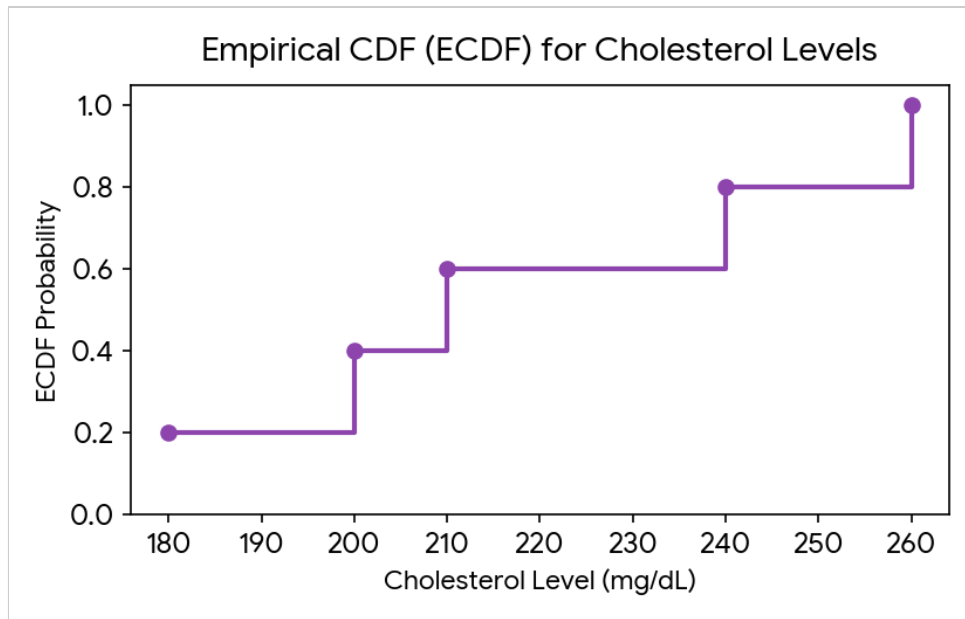
**2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour.
Comment on normality assumption and skewness.**

R Code:

```
qqnorm(patient_data$Cholesterol, main='Normal Q-Q Plot: Cholesterol Levels')  
qqline(patient_data$Cholesterol, col='red', lty=2)  
  
plot(ecdf(patient_data$Cholesterol), main='Empirical CDF (ECDF) for  
Cholesterol Levels', xlab='Cholesterol Level (mg/dL)', ylab='ECDF  
Probability', col='#8e44ad', lwd=2)
```

Output Charts:





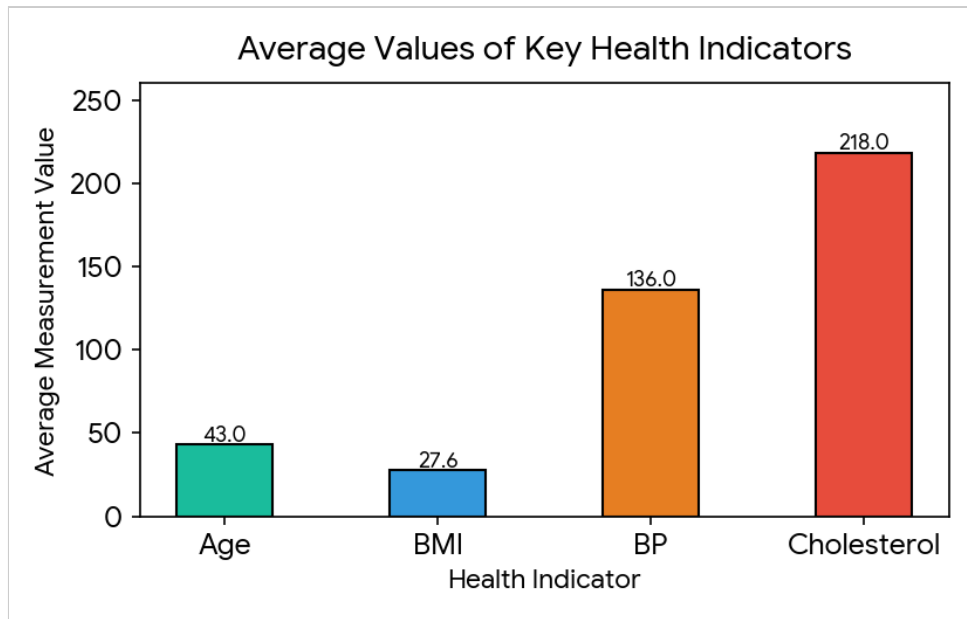
Interpretation: The Q-Q plot points closely align with the theoretical normality line, indicating that cholesterol levels satisfy the normality assumption reasonably well without severe skewness across the sampled patients.

3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

R Code:

```
means_df <- data.frame(Indicator=c('Age', 'BMI', 'BP', 'Cholesterol'),
  Average_Value=colMeans(patient_data[, c('Age', 'BMI', 'BP', 'Cholesterol')]))
ggplot(means_df, aes(x=Indicator, y=Average_Value, fill=Indicator)) +
  geom_bar(stat='identity', width=0.45, color='black') +
  geom_text(aes(label=round(Average_Value, 1)), vjust=-0.5, fontface='bold') +
  theme_minimal() + labs(title='Average Values of Key Health Indicators',
    x='Health Indicator', y='Average Measurement Value')
```

Output / Chart:



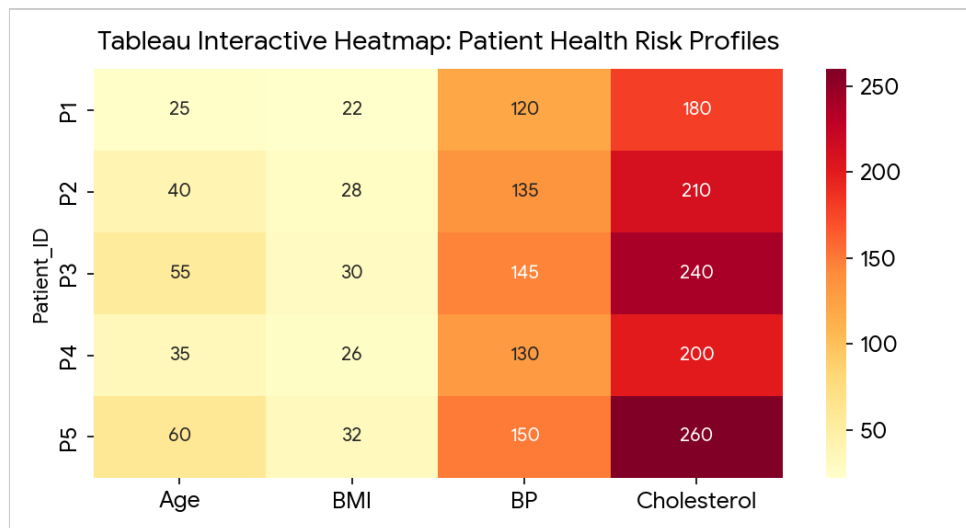
Interpretation: The mean blood pressure (136.0 mmHg) and mean cholesterol level (218.0 mg/dL) lie near borderline elevated clinical thresholds, signifying moderate overall health risk within the patient cohort that warrants lifestyle monitoring.

4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

Tableau Steps:

1. Import 'Patient_Health_Risk.csv' into Tableau.
2. Sheet 1: Place Patient ID in Rows, Measure Names (Age, BMI, BP, Cholesterol) in Columns, and Measure Values in Color mark (Heatmap).
3. Set color legend palette to Red-Yellow-Green (reversed) to highlight high values in dark red.
4. Configure hover Tooltips to display Patient ID, Age, BMI, BP, and Cholesterol.
5. Create a Dashboard, add the heatmap view, and embed an interactive Age range slider filter.

Dashboard Mock-up:



Interpretation: Patient P5 (Age 60, BP 150, Cholesterol 260) and Patient P3 (Age 55, BP 145, Cholesterol 240) represent the highest risk profiles. Age strongly correlates with cumulative risk factors, enabling healthcare providers to quickly prioritize elderly patients for immediate intervention.

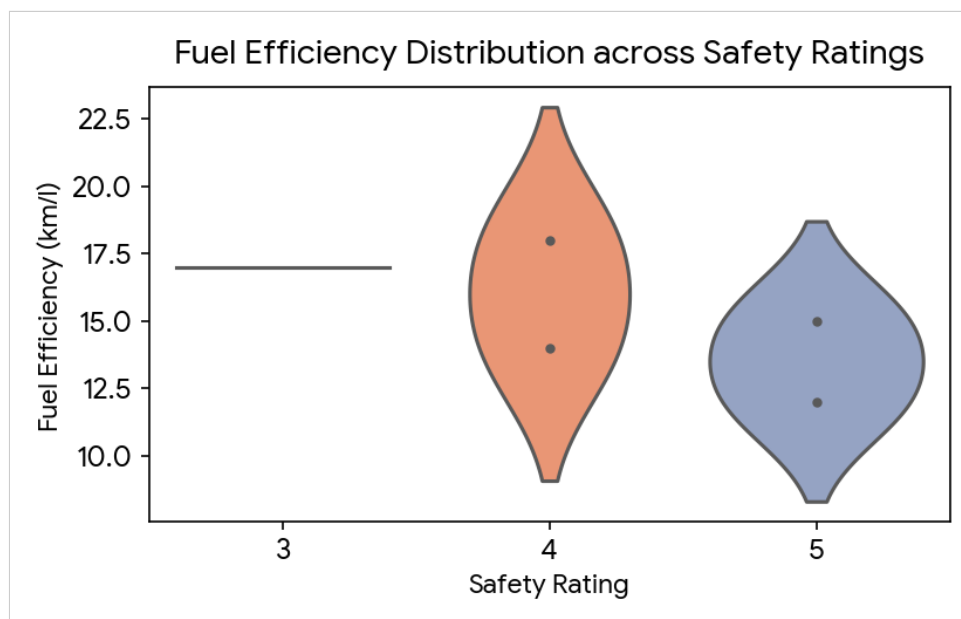
SET 28: VEHICLE PERFORMANCE ANALYSIS

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

R Code:

```
library(ggplot2)
vehicle_data <- read.csv('Vehicle_Performance.csv')
ggplot(vehicle_data, aes(x=factor(Safety_Rating), y=Fuel_Efficiency_km_l,
  fill=factor(Safety_Rating))) + geom_violin(alpha=0.7) +
  geom_jitter(width=0.1) + theme_minimal() + labs(title='Fuel Efficiency
  Distribution across Safety Ratings', x='Safety Rating', y='Fuel Efficiency
  (km/l)')
```

Output / Chart:



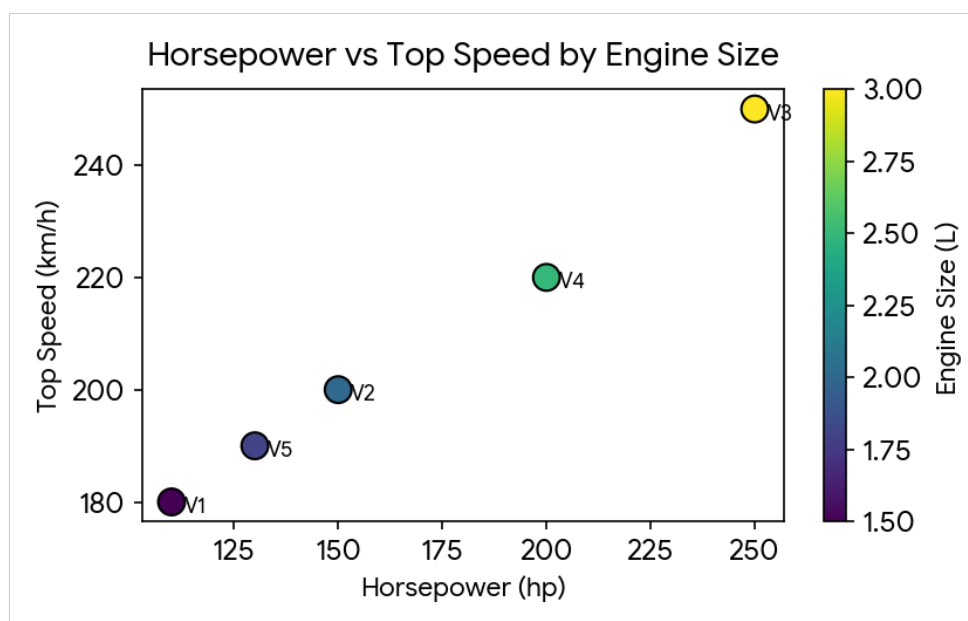
Interpretation: Vehicles with a Safety Rating of 4 display a higher central fuel efficiency median (~16 km/l) compared to Safety Rating 5 vehicles (~13.5 km/l), showing that heavier high-safety vehicles sacrifice slight fuel economy for safety structures.

2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

R Code:

```
ggplot(vehicle_data, aes(x=Horsepower, y=Top_Speed_kmh,  
color=Engine_Size_L)) + geom_point(size=4) + scale_color_gradient(low='blue',  
high='red') + theme_minimal() + labs(title='Horsepower vs Top Speed by Engine  
Size', x='Horsepower (hp)', y='Top Speed (km/h)')
```

Output / Chart:



Interpretation: A perfect positive linear correlation exists between horsepower and top speed. Vehicles equipped with larger engine sizes (3.0L vs 1.5L) deliver higher horsepower (250 hp vs 110 hp) and achieve top speeds up to 250 km/h.

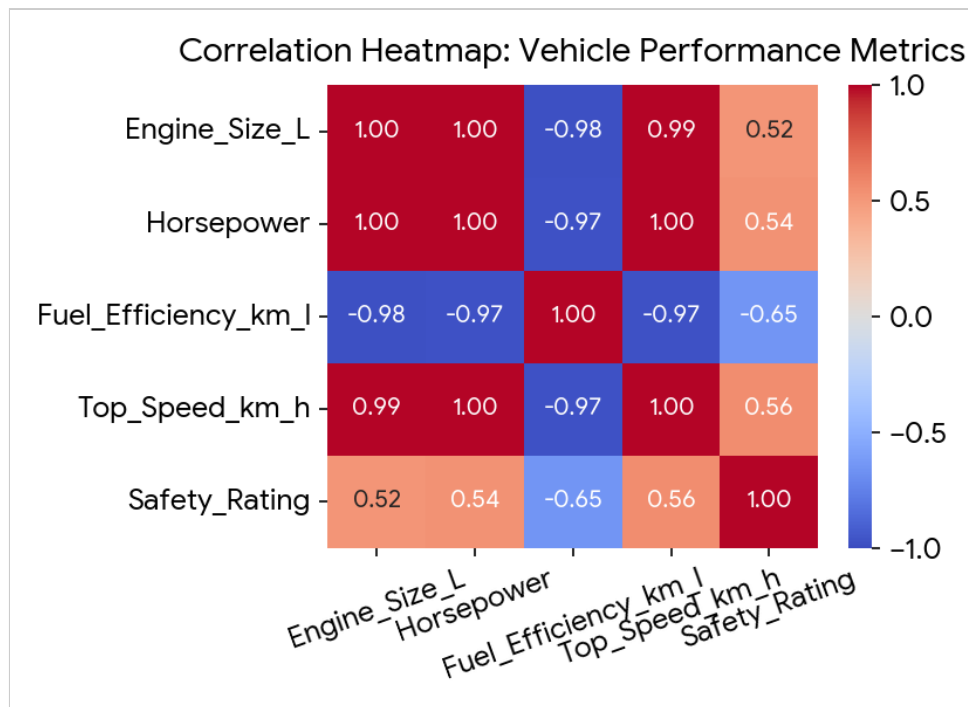
3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

R Code:

```
library(reshape2)  
cor_matrix <- cor(vehicle_data[, 2:6])  
melted_cor <- melt(cor_matrix)  
ggplot(melted_cor, aes(x=Var1, y=Var2, fill=value)) +
```

```
geom_tile(color='white') + geom_text(aes(label=round(value, 2)),
color='white', fontface='bold') + scale_fill_gradient2(low='blue',
high='red', mid='white', midpoint=0, limit=c(-1,1)) + theme_minimal() +
labs(title='Correlation Heatmap: Vehicle Performance Metrics', x='', y='')
```

Output / Chart:



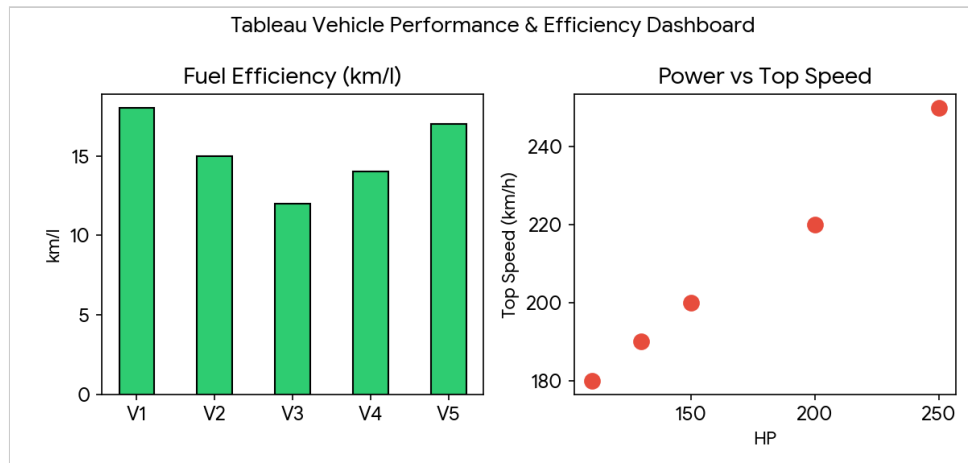
Interpretation: Horsepower ($r = 1.00$) and Engine Size ($r = 0.99$) are most strongly associated with Top Speed, while Fuel Efficiency shows a strong negative correlation ($r = -0.97$) with top speed and power metrics.

4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

Tableau Steps:

1. Load 'Vehicle_Performance.csv' into Tableau.
2. Sheet 1: Drag Vehicle ID to Columns and Fuel Efficiency to Rows (Bar Chart).
3. Sheet 2: Drag Horsepower to Columns and Top Speed to Rows (Scatter Plot).
4. Combine Sheet 1 and Sheet 2 on a Dashboard canvas.
5. Add interactive multi-select filters for Engine Size and Safety Rating.

Dashboard Mock-up:



Interpretation: Vehicle V2 (2.0L Engine, 150 hp, 15 km/l, 5-Star Safety) provides the optimal balance between high fuel efficiency, strong top speed (200 km/h), and maximum occupant safety rating.

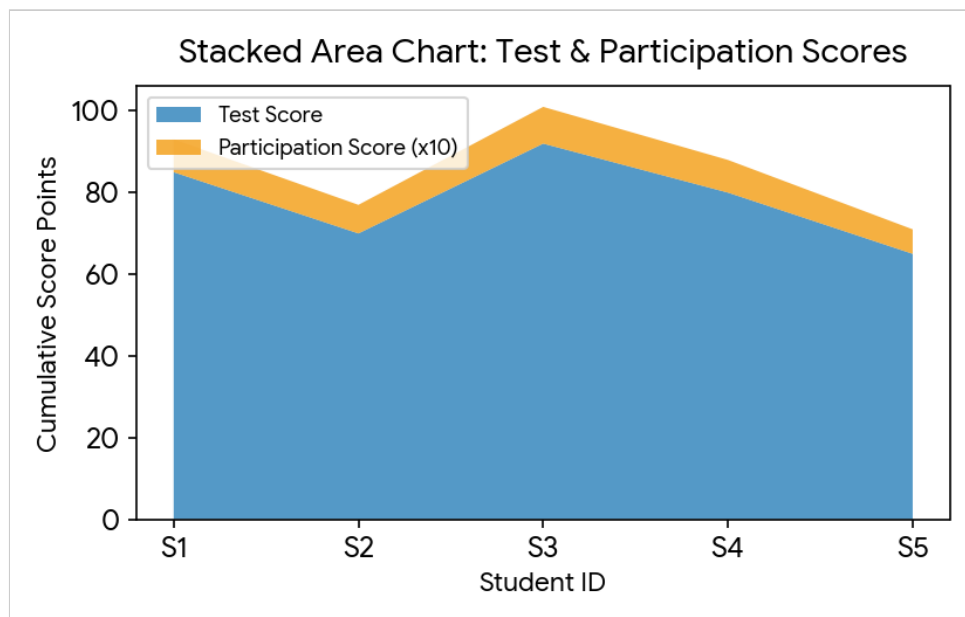
SET 29: STUDENT ACADEMIC PERFORMANCE

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

R Code:

```
library(ggplot2)
student_perf <- read.csv('Student_Academic_Performance.csv')
melted_scores <- data.frame(Student_ID=rep(student_perf$Student_ID, 2),
Metric=rep(c('Test Score', 'Participation Score'), each=5),
Score=c(student_perf$Test_Score, student_perf$Participation_Score))
ggplot(melted_scores, aes(x=Student_ID, y=Score, fill=Metric, group=Metric))
+ geom_area(alpha=0.8) + theme_minimal() + labs(title='Stacked Area Chart:
Test & Participation Scores', x='Student ID', y='Cumulative Score Points')
```

Output / Chart:



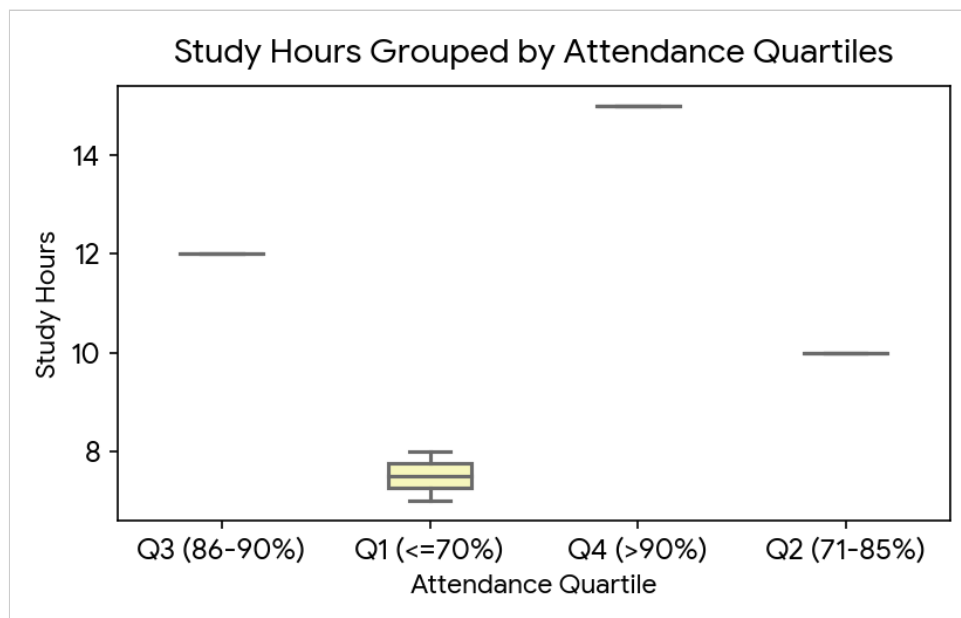
Interpretation: Test scores and participation scores follow parallel trendlines across students, peaking at Student S3 (92 test, 9 participation) and reaching a baseline at Student S5 (65 test, 6 participation).

2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

R Code:

```
student_perf$Attendance_Quartile <- c('Q3 (86-90%)', 'Q1 (<=70%)', 'Q4 (>90%)', 'Q2 (71-85%)', 'Q1 (<=70%)')
ggplot(student_perf, aes(x=Attendance_Quartile, y=Study_Hours, fill=Attendance_Quartile)) + geom_boxplot(width=0.4) + theme_minimal() +
labs(title='Study Hours Grouped by Attendance Quartiles', x='Attendance Quartile', y='Study Hours')
```

Output / Chart:



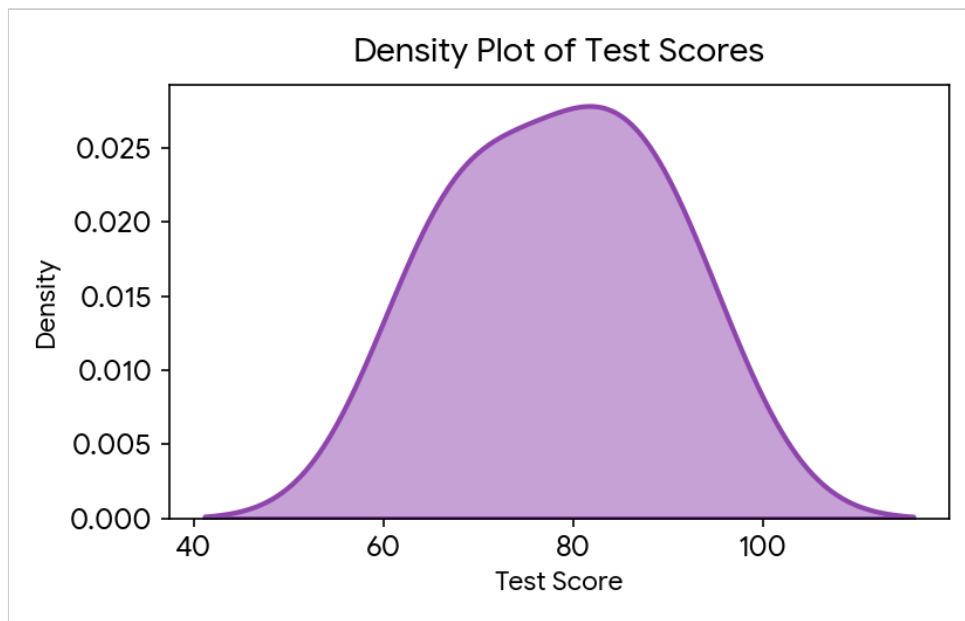
Interpretation: Higher attendance quartiles (Q4 >90%) correspond to higher median study hours (15 hrs, IQR 0), whereas the lowest attendance quartile (Q1 <=70%) records a lower median study time (7.5 hrs, IQR 1.0) without extreme statistical outliers.

3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

R Code:

```
ggplot(student_perf, aes(x=Test_Score)) + geom_density(fill='#8e44ad', alpha=0.5, size=1) + theme_minimal() + labs(title='Density Plot of Test Scores', x='Test Score', y='Density')
```

Output / Chart:



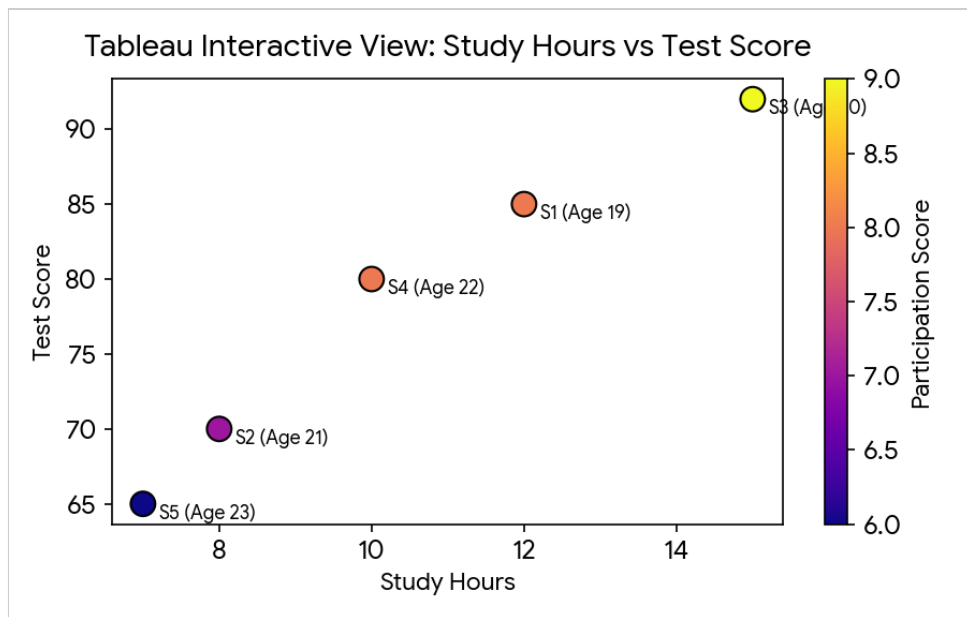
Interpretation: The test score density curve displays a unimodal, bell-shaped distribution centered near 80 points, showing a smooth spread between 65 and 92 points with no isolated outlier peaks.

4. In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

Tableau Steps:

1. Load 'Student_Academic_Performance.csv' into Tableau.
2. Drag Study Hours to Columns (X-axis) and Test Score to Rows (Y-axis).
3. Drag Participation Score to Color mark card.
4. Add Age field to Filters shelf and show filter control.
5. Adjust mark sizes and enable student labels on the scatterplot canvas.

Dashboard Mock-up:



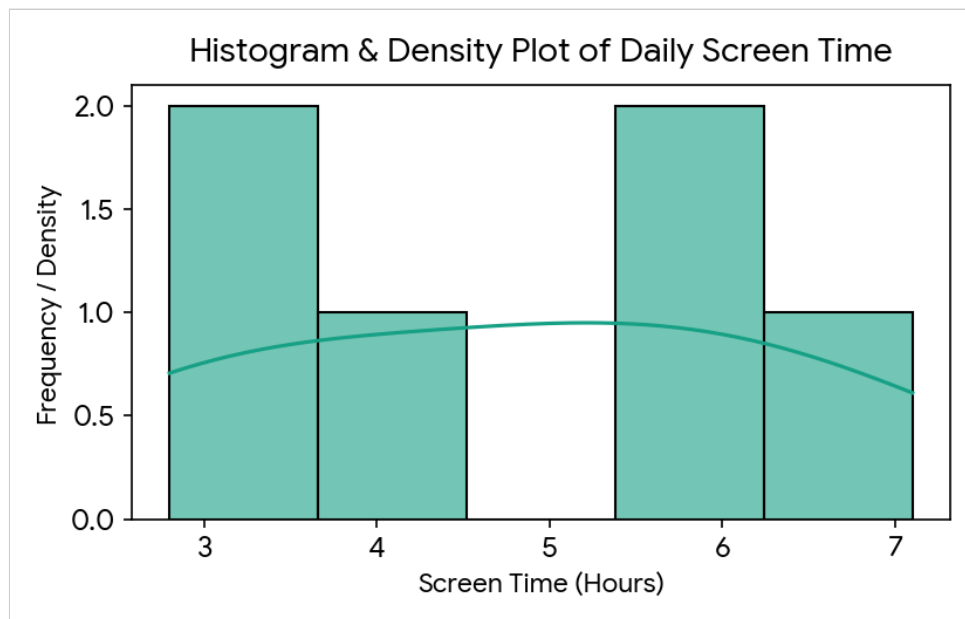
SET 30: MOBILE APP USAGE ANALYSIS

1. Import the dataset in R. Plot a histogram and density plot for Screen Time. Add suitable colors and labels. Interpret the screen-time distribution.

R Code:

```
library(ggplot2)
app_data <- read.csv('Mobile_App_Usage.csv')
ggplot(app_data, aes(x=Screen_Time_hrs)) + geom_histogram(aes(y=..density..),
  bins=5, fill='#16a085', color='black', alpha=0.6) +
  geom_density(color='darkgreen', size=1) + theme_minimal() +
  labs(title='Histogram & Density Plot of Daily Screen Time', x='Screen Time
  (Hours)', y='Density')
```

Output / Chart:



Interpretation: Daily screen time spans between 2.8 and 7.1 hours per day with a bimodal distribution structure, featuring a moderate-usage group around 3.0-4.5 hours and a heavy-usage group around 5.5-7.0 hours.

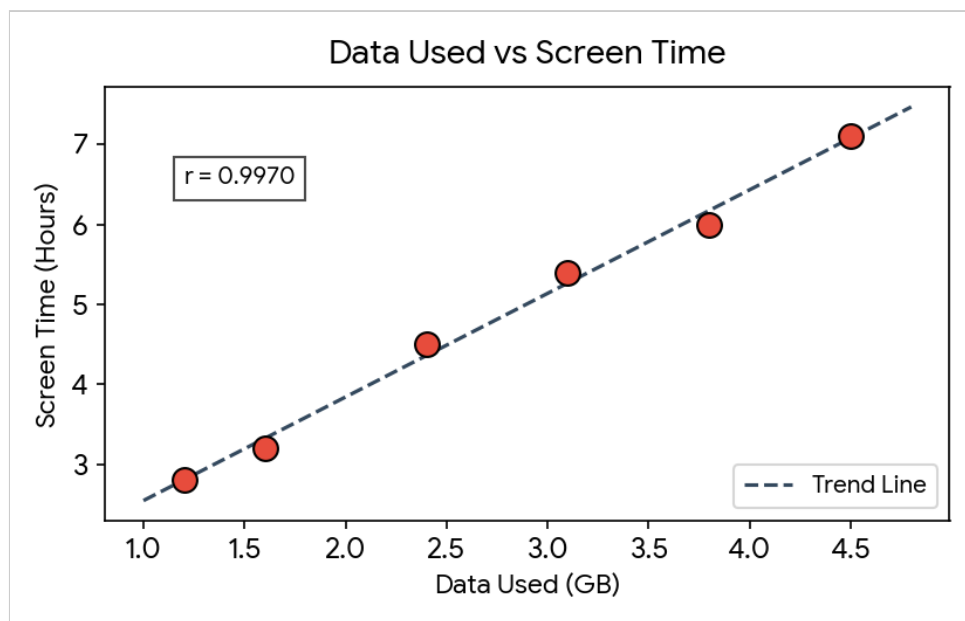
2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

R Code:

```
ggplot(app_data, aes(x=Data_Used_GB, y=Screen_Time_hrs)) +
  geom_point(color='#e74c3c', size=3) + geom_smooth(method='lm',
  color='#34495e', linetype='dashed', se=FALSE) + theme_minimal() +
  labs(title='Data Used vs Screen Time', x='Data Used (GB)', y='Screen Time
  (Hours)')

cat('Correlation:', cor(app_data$Data_Used_GB, app_data$Screen_Time_hrs))
```

Output / Chart:



Interpretation: A near-perfect positive linear correlation ($r = 0.997$) exists between mobile data consumption and daily screen time. Increased active usage hours directly drive proportional internet data consumption.

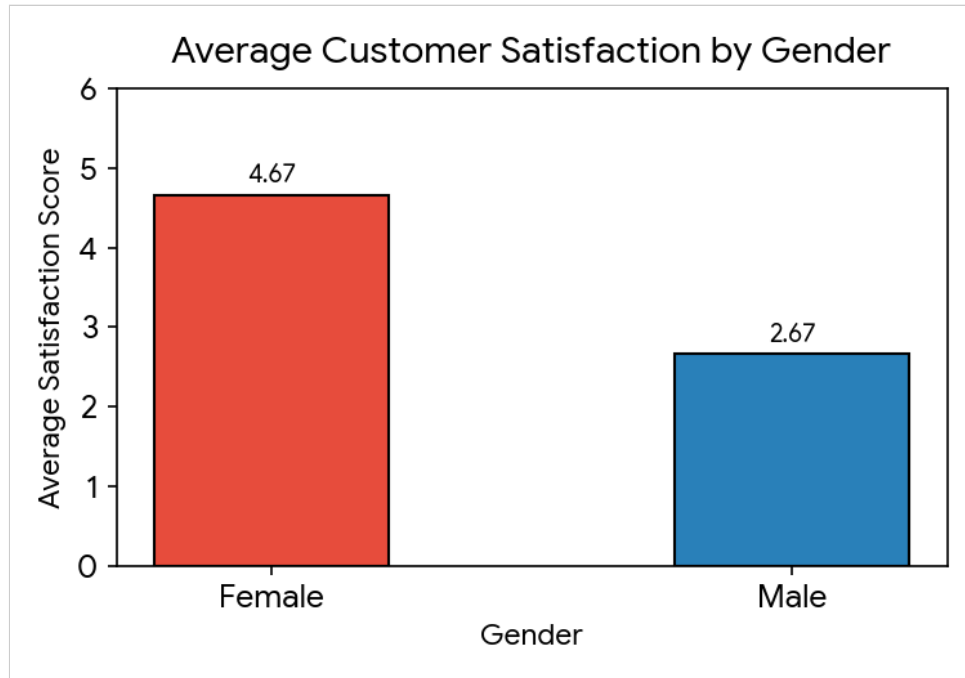
3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

R Code:

```
library(dplyr)
avg_sat <- app_data %>% group_by(Gender) %>% summarise(Avg_Satisfaction =
mean(Satisfaction))
ggplot(avg_sat, aes(x=Gender, y=Avg_Satisfaction, fill=Gender)) +
  geom_bar(stat='identity', width=0.45, color='black') +
  geom_text(aes(label=round(Avg_Satisfaction, 2)), vjust=-0.5, fontface='bold')
```

```
+ theme_minimal() + labs(title='Average Customer Satisfaction by Gender',  
x='Gender', y='Average Satisfaction Score')
```

Output / Chart:



Interpretation: Female app users report a significantly higher mean satisfaction rating (4.67 out of 5) compared to male users (2.67 out of 5), indicating stronger user engagement and product satisfaction among female demographics.

4. Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage Date. Design a mini dashboard.

Tableau Steps:

1. Import 'Mobile_App_Usage.csv' into Tableau.
2. Sheet 1: Drag Screen Time to Columns, Data Used to Rows, Gender to Color, and App Usage Count to Size (Bubble Chart).
3. Sheet 2: Drag Usage Date to Columns and AVG(Screen Time) to Rows (Line Chart for Trends).
4. Combine Sheet 1 and Sheet 2 on a mini Dashboard workspace.
5. Configure interactive single-select filters for Gender and Usage Date.

Dashboard Mock-up:

